

Ahmed Abdelrahman

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Skills

languages TypeScript, Rust, Python	frontend React, Next.js, Tailwind CSS	backend Node.js, Express.js, REST APIs, Event-Driven Architecture
databases PostgreSQL, MongoDB, Redis	infrastructure AWS (ECS Fargate, ECR, RDS, ElastiCache, Lambda, SSM), Docker, GitHub Actions CI/CD, Vercel	core Distributed Systems, System De- sign, Database Internals, Clean Code & Scalability

Education

Faculty of Pharmacy 2024 Modern University, Cairo, Egypt

Work Experience

Vogel Edits 01/2025 - Present
Full-Stack Engineer (Adobe Extensions)

- Built and shipped production Adobe After Effects & Premiere Pro extensions, including Vanish BG — a real-time AI background removal tool available at aescrpts.com, running ML inference inside a sandboxed CEP environment.
- Designed a Node.js sidecar process architecture to handle AI inference outside Adobe's restricted CEP sandbox, using local HTTP for IPC and streaming results back to the React UI in real time.
- Built a TTS Premiere Pro extension using Node.js with local ONNX model inference, designing the full audio pipeline from text input to rendered timeline clip.
- Collaborated asynchronously with designers and stakeholders, shipping production-ready tools using agile workflows.

Freelance 11/2024 - 01/2025
Full-Stack Engineer

- Built a full-stack tourism booking platform using Next.js, TypeScript, and Node.js with TypeORM for type-safe PostgreSQL access.
- Applied PostgreSQL indexing strategies and EXPLAIN ANALYZE to achieve sub-100ms API response times under concurrent load.
- Integrated Paymob payment gateway with idempotent transaction handling and webhook verification, reducing checkout friction by 40%.
- Improved SEO and Core Web Vitals to 95+ Lighthouse score using Next.js SSR, image optimization, and lazy loading.

Projects

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>
Tech Stack: TypeScript, Node.js, PostgreSQL, Redis, AWS, Docker, BullMQ

- Designed and deployed a distributed log ingestion pipeline on AWS ECS Fargate processing 1M+ events/hour with sub-second latency.
- Reduced ONNX embedding time from 20+ minutes to under 1 minute via batch processing, Redis caching, and worker autoscaling.

- Designed an event-driven pipeline — normalize, embed, cluster, merge, root cause — with retries, exponential backoff, DLQs, and idempotent processing.
- Set up full CI/CD with GitHub Actions OIDC, ECR, SSM Parameter Store, RDS PostgreSQL, and ElastiCache.
- Applied PostgreSQL internals (indexing, MVCC, isolation levels) to build a high-throughput write pipeline avoiding lock contention.

Medical CRM — AI-Enhanced Prescription Platform

Tech Stack: React, Node.js, PostgreSQL, JWT, RBAC

- Built a full-stack CRM handling diagnoses, prescriptions, and patient records with React, Node.js, and PostgreSQL.
- Implemented JWT auth with access/refresh token rotation and RBAC across three permission levels.
- Designed drug interaction validation across 500+ medications using graph-based PostgreSQL lookup.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.